

Constraint-Aware Interaction Generator for Recommender System Simulation

Master's Thesis Research Plan, Master of Artificial Intelligence

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Recommender modeling has broadly adopted transformer architectures for sequential tasks, while synthetic interaction generation in recommender simulation remains dominated by process-based and adversarial approaches. This thesis addresses that gap with a hybrid architecture: a multi-head autoregressive transformer embedded in a process-based simulation wrapper, generating typed event sequences with explicit hard constraints on action transitions and temporal consistency.

The scope is a comparison between the proposed transformer generator against a Markov chain baseline, evaluated on the Synerise RecSys 2025 dataset across three dimensions: fidelity, validity, and downstream utility. The intended output is a working interaction generator and a reusable evaluation methodology for simulator quality claims in recommender-system research.

Additional Key Words and Phrases: recommender systems, synthetic data, simulation, transformer, interaction generation, evaluation protocol

1 Introduction and Problem Context

Recommender systems (RS) are frequently developed and evaluated on static historical logs. This is practical but methodologically incomplete. When recommendation policies change, user exposure changes. When exposure changes, user behavior changes. When behavior changes, future training data changes. Static snapshots therefore fail to capture the feedback dynamics that emerge in iterative deployment. Offline evaluation on fixed logs can look convincing while still missing important behavioral effects that only materialize under continuous operation. This challenge is repeatedly identified in the recommender simulation literature as a fundamental reason for interest in synthetic interaction generation as a methodological tool [1–3].

The RS simulation literature is active but methodologically fragmented. Existing approaches include process-based and framework-oriented simulators [4–6], adversarial and statistical generators [7, 8], and configurable data synthesis tools [13, 14]. These lines provide important building blocks but also reveal recurring issues: heterogeneous assumptions, inconsistent metric usage, and weak comparability across studies [1, 3]. At the same time, sequential recommendation has demonstrated strong performance from transformer architectures, with SASRec [15], BERT4Rec [16], and GRU4Rec [17] representing the progression from recurrent to attention-based methods. However, prediction success does not automatically transfer to synthetic interaction generation. Generator design requires distinct choices around output structure, validity constraints, temporal handling, and evaluation criteria.

The present thesis is positioned directly in this gap. The project context is a two-component recommender system simulator: an arrival component (component 1) is already operational, while an interaction generator (component 2) is the open problem. The prior implementation couples a first-order Markov chain with an LSTM that functions as a feature extractor for item ranking, not a learned sequence generator. The interaction generation is therefore a patchwork of three mechanisms: a recommender that selects items, a Markov chain that picks behavior transitions, and hardcoded session logic that glues them together. There is no single learned model generating coherent interaction sequences end-to-end. The result is implausible session structures: action

sequences that violate behavioral funnels at rates incompatible with observed data, and insufficient distributional coverage across the item space.

This thesis bridges those two lines: a transformer-backed interaction generator embedded in a process-based simulation wrapper, evaluated under falsifiable criteria for fidelity, validity, and utility. The contribution is both technical, in the form of a multi-head autoregressive architecture for typed event tuple generation, and methodological: a reproducible evaluation protocol for simulator quality claims, addressing the evaluation fragmentation documented across the reviewed corpus.

2 Research Goal and Questions

2.1 Research Goal

The goal is to design and validate a constraint-aware interaction generator that produces synthetic interaction logs suitable for recommender-system training and evaluation. The contribution is:

- **Technical:** a hybrid architecture combining a process-based simulation wrapper with a multi-head autoregressive transformer as the interaction generation backbone;
- **Methodological:** a reproducible and falsifiable evaluation protocol for simulator quality, covering fidelity, validity, and downstream utility.

2.2 Main Research Question

How and to what extent can a constraint-aware autoregressive transformer interaction generator produce synthetic interaction logs with high fidelity, validity, and downstream utility for recommender-system training and evaluation?

2.3 Research Sub-questions

The main question decomposes into four sub-questions, each mapping to one of the assignment’s research questions (a–d):

- (1) **SQ1:** How should interactions be formally defined for simulation purposes, covering event fields, temporal structure, and session dependencies?
- (2) **SQ2:** What architectural properties does an interaction generator need to capture realistic interaction dynamics, including sequential dependence, action-item coherence, and temporal structure?
- (3) **SQ3:** Which generator implementations and baseline policy are feasible within thesis scope for a fair and executable comparison?
- (4) **SQ4:** Which evaluation protocol makes “better than baseline” falsifiable and reproducible?

3 Stakeholder Analysis

3.1 Stakeholders

Academic supervisor (Diptish Dey, CMI). Diptish Dey is the primary owner of this project. His stake is in a working, validated interaction generator that integrates with the existing arrival simulator. Timely delivery and a defensible evaluation methodology matter to him because the simulator is part of ongoing research at CMI.

CMI research group. The broader CMI research group has a practical stake in the technical output. Colleagues working on related projects stand to benefit from a reusable simulator component and a documented evaluation protocol they can apply or extend without having to rebuild from scratch.

Recommender systems research community. The research community’s interest is in reusable evaluation protocols, open-dataset compatibility, and reproducible baselines that support cumulative scientific progress.

Dataset provider (Synerise / RecSys 2025 challenge). Synerise’s interest is in demonstrating the richness and utility of their publicly released dataset. As the dataset is publicly released under challenge terms, Synerise has no active supervisory or evaluative role in this project.

Downstream users of the recommender system. The behavioral patterns in the Synerise logs represent real user populations. These anonymous users have an interest in synthetic data not amplifying distributional biases present in the real logs, such as popularity skew. This interest is addressed through bias metrics (Jensen-Shannon divergence on action distributions, item coverage) reported as part of the fidelity evaluation. The Synerise logs represent real people’s shopping behavior; this research neither re-identifies nor profiles any individual in accordance with the legal requirements detailed in Section 6.

3.2 Conflicting Interests and Prioritization

The stakeholder interests above are largely aligned but diverge on two points.

The first tension is between the academic supervisor’s interest in timely delivery and the research community’s interest in broad generalizability. A more extensive comparison portfolio, covering additional baselines such as GRU or GAN-based generators, would strengthen the community contribution but risks overrunning the 14-week thesis timeline. This tension is resolved in favor of the supervisor’s constraint: scope is fixed to a single transformer versus Markov chain comparison, with extensions explicitly identified as future work (Section 11).

The second tension is between Synerise’s interest in showcasing dataset richness and the downstream users’ interest in not amplifying bias. A generator optimized purely for distributional fidelity to the training logs will faithfully reproduce whatever popularity skew and temporal selection effects are present in the Synerise data. Downstream users are better served by a generator that surfaces and reports those biases rather than silently inheriting them. This tension is resolved by including explicit bias metrics in the fidelity evaluation rather than treating fidelity-to-data and fairness-to-users as equivalent goals.

4 State of Field and Literature Gap

4.1 Recommender System Simulation Landscape

The RS simulation literature is active but methodologically fragmented. Stavinova et al. [1] document heterogeneous simulators, missing standardized head-to-head comparisons, and weak universal quality-control methodologies. SimuRec [2] identifies the absence of shared standards as a community-level gap and motivates the need for more systematic evaluation frameworks. Heterogeneous evaluation practices mean that results across existing simulators are rarely directly comparable, undermining cumulative scientific progress.

Process-based and framework-oriented simulators form one branch of the literature. SIREN [4] demonstrates the simulation of longitudinal recommender effects in online news environments, with explicit user-system feedback dynamics modeled over time. T-RECS [5] lowers the barrier to long-horizon RS simulation through a modular, agent-based design and multi-run confidence intervals for variance-aware analysis. Accordion [6] models user visits as inhomogeneous Poisson processes with marked events, demonstrating that explicit temporal process modeling substantially changes simulated long-term interactive effects. These tools establish the system-level context this thesis operates within, while leaving event-level interaction generation largely to domain assumptions or simplified behavioral rules.

Adversarial and statistical generation form a second branch. UserSim [7] applies a supervised GAN to simulate user behavior for offline reinforcement learning over recommender systems. Bobadilla et al. [8] generate parameterized collaborative filtering datasets via GAN for configurable

benchmark construction. DataGenCARS [13] and Jakomin et al. [14] provide configurable statistical generators for context-aware and stream-based interaction data. These show the range of non-transformer approaches and their practical tradeoffs. Chaney [3] explicitly critiques simulation assumptions and the risk of overconfident conclusions without explicit validity checks, a concern this thesis addresses directly.

Evaluation challenges are a cross-cutting concern. Figueira and Vaz [10] identify that single-metric evaluation is insufficient for synthetic data claims. Jordon et al. [11] articulate that utility-only validation does not guarantee distributional fidelity. Lu et al. [12] identify missing standardized evaluation tools across the synthetic data generation field. This convergent critique motivates the multi-axis evaluation design of this thesis.

4.2 Sequential Modeling and Transformers

Transformers have proven dominant for next-item prediction in sequential recommendation. SAS-Rec [15] established causal self-attention for sequential RS, demonstrating that attention-based models substantially outperform Markov chains and RNNs on next-item prediction benchmarks. BERT4Rec [16] extended this with masked training objectives for bidirectional context modeling. GRU4Rec [17] is the canonical RNN predecessor that transformers have consistently outperformed on longer sequences.

However, next-item prediction and synthetic interaction log generation are distinct tasks. Prediction asks: given history, what item comes next? Generation asks: produce a full, distributionally realistic interaction session from scratch. The output structure, evaluation criteria, and success measures differ fundamentally. This thesis applies transformer sequence modeling to generation, not prediction; the architectural transfer is strongly motivated by evidence but cannot be assumed automatic. Generated logs require additional evaluation along fidelity and validity axes that next-item accuracy alone does not capture.

4.3 The Intersection Gap

Across the RS simulation literature reviewed for this thesis, no work that uses a non-LLM autoregressive transformer as the primary architecture for generating enriched interaction event logs was identified. The RS simulation branch relies entirely on process-based, agent-based, statistical, or adversarial approaches, while the sequential recommendation branch applies transformers exclusively to next-item prediction, not to session generation. This work positions itself at this intersection: applying transformer sequence strengths to interaction generation, embedded in a process-based simulation wrapper, and evaluated with a unified protocol that has not been applied to this generator family in the reviewed literature.

5 Theoretical Foundation

The previous section mapped the landscape of existing work. This section draws on that literature selectively to derive the theoretical commitments that ground the design decisions in this thesis. Each subsection identifies a principle about interaction structure, modeling assumptions, evaluation, or automation and shows how it motivates architectural and methodological choices.

5.1 Interaction as Typed Temporal Event Sequences

Across the RS simulation papers reviewed for this thesis, “interaction” is more than a $\langle \text{user}, \text{item}, \text{rating} \rangle$ tuple. Accordion models events as $(\text{time}, \text{user}, \text{item}, \text{outcome})$ with visit dynamics [6]. UserSim models sequential user behavior conditioned on prior interactions [7]. SIREN and T-RECS model iterative, stateful user-item processes over time [4, 5]. DataGenCARS formalizes context as a generated part of the interaction record [13]. This convergence of evidence across different

simulator families supports a two-level interaction definition: event-level typed tuples grouped into sessions, with user state carried across sessions via history. This definition directly motivates the multi-head prediction architecture, as each field of the tuple becomes a separate conditional prediction target.

5.2 Sequential Dependence as a Modeling Requirement

Realistic interaction sequences are not independent and identically distributed events. A user who views an item, then adds it to cart, then removes it, exhibits session-level dependence that first-order Markov chains cannot capture reliably. Chaney [3] identifies missed behavioral dependencies as a central simulation validity risk, particularly when models make implicit IID assumptions that the real data does not support. Accordion demonstrates that temporal process modeling changes long-term simulation effects substantially [6]. This motivates sequence-capable generators over IID-only approaches and justifies the autoregressive design of the proposed transformer.

5.3 Adversarial Generation: Utility and Risk Tradeoffs

GAN-based user simulators provide utility-oriented evidence [7, 8] but carry well-documented limitations. Creswell et al. [9] document mode collapse and optimization instability as structural GAN risks, while Figueira and Vaz [10] note persistent evaluation ambiguity. For discrete sequential generation specifically, gradient flow through discrete tokens requires additional workarounds that compound architectural complexity. Autoregressive transformers with softmax output heads handle discrete token generation natively, which is a structural advantage for the interaction generation task.

5.4 Multi-Axis Evaluation as a Scientific Requirement

Jordon et al. [11] argue that utility-only validation is insufficient: synthetic data that enables good downstream models may still be statistically implausible at the distributional level. Figueira and Vaz [10] reinforce that single-metric evaluation is systematically incomplete. Stavinova et al. [1] document heterogeneous and non-comparable metrics across simulators. Taken together, these findings make a case for evaluating fidelity, validity, and utility jointly.

5.5 Reproducibility and Assumption Explicitness

T-RECS emphasizes multi-run confidence intervals and repeated-run variance analysis as scientific requirements for simulation research [5]. SimuRec identifies reproducibility as a missing standard across the community [2]. Chaney argues that simulation conclusions are only credible when modeling assumptions are stated explicitly and tested empirically [3]. This thesis operationalizes these requirements through fixed random seeds, deterministic preprocessing, and a hard-constraint validity layer that makes generation assumptions explicit and measurable. Results are reported across five seeds to surface variance.

5.6 AI Suitability and Degree of Automation

Rule-based approaches to interaction generation, such as handcoded Markov chains with manually specified legal transitions, require exhaustive elicitation of every valid behavioral path and cannot generalize to unseen contexts or long-range session dependencies. Learned models extract transition probabilities from data and capture dependencies that fixed rules cannot express [3, 6]. The case for a learned model over handcoded rules follow from this.

Autoregressive (sequential) generation is not the only design option for learned sequence models. Parallel or non-autoregressive generation methods produce all output tokens simultaneously, offering higher throughput. However, parallel generation requires conditional independence assumptions across output positions that are incompatible with the behavioral dependencies documented in the literature: a generated action type must condition on prior actions, and a generated item must condition on the current action type. Violating these dependencies at generation time would undermine the validity of the constraint layer and the fidelity of the output log [3]. Autoregressive generation is therefore the appropriate choice for this task, with throughput as an acknowledged tradeoff documented in Risk 5.

Among learned approaches, a transformer architecture is preferred over simpler sequence models. The factored multi-head design enables joint action-item-time modeling, and self-attention captures long-range session dependencies more effectively than first-order Markov chains or other recurrent architectures [15–17].

Full automation of the generation pipeline is appropriate for this use case because the output is synthetic training data rather than decisions affecting real users directly. No human-in-the-loop is required at generation time. However, the hard-constraint validity layer makes the automation explicit and auditable: generation assumptions are documented as constraint rules extracted from real data, and violations are counted and reported.

The broader societal implications of this automation warrant explicit consideration. Recommender systems trained on synthetic interaction data influence what products, content, and services real users are exposed to at scale. If the generator systematically amplifies distributional biases present in the training data — such as popularity skew toward dominant items or underrepresentation of niche user behaviors — downstream recommenders trained on that data may entrench those patterns further. Users whose behavior falls outside the generator’s learned distribution risk receiving lower-quality recommendations, potentially reinforcing existing inequalities in exposure and access.

The most significant risk is feedback loop amplification: a biased generator produces biased training data, which trains a biased recommender, which produces biased interaction logs, which are used to retrain the generator. Each cycle compounds the original distortion. This risk is not fully neutralized by bias reporting alone — measuring a problem does not resolve it. The mitigation in this thesis is therefore scoped honestly: bias metrics are reported to make the problem visible and quantifiable, enabling downstream researchers and practitioners to make informed decisions about whether and how to deploy synthetic data produced by this pipeline.

Given these risks, the research is nonetheless justified. The alternative, continued reliance on static historical logs for recommender evaluation, carries its own unacknowledged biases and feedback dynamics. A transparent, auditable synthetic generator with documented bias characteristics is a more controllable research instrument than an unexamined historical log. The contribution of this thesis is therefore not a risk-free tool, but a tool whose risks are made explicit, which is a precondition for responsible use.

6 Requirements

6.1 Legal Requirements

The primary dataset is the Synerise RecSys 2025 dataset, released publicly under the RecSys 2025 challenge terms. The dataset consists of anonymized behavioral event logs with no personally identifiable information (PII). Processing anonymized behavioral data for academic research is permissible under the scientific research exemption of the EU General Data Protection Regulation (GDPR, Art. 89) [18]. Generated synthetic data carries no re-identification risk: item-ID vocabulary

capping reduces the item space to a top- K vocabulary, and all outputs are population-level statistics rather than individual records.

6.2 Ethical Requirements

Synthetic interaction data trained on real behavioral logs inherits the distributional properties of that data, including popularity skew, temporal selection effects, and potential demographic proxy signals [3, 11]. This risk is acknowledged and is made visible by reporting bias metrics as part of the fidelity evaluation: Jensen-Shannon divergence over the action-type distribution and item coverage are both reported. Reporting does not eliminate the inherent bias in the training distribution, but does make the inherited skew transparent.

The generation pipeline must not be used for individual-level profiling or decisions. Use is restricted to aggregate training data generation. The explicit hard-constraint layer documents and enforces generation assumptions, making the automation auditable.

6.3 Organizational Requirements

The project is conducted by a single developer within a 14-week thesis timeline. The deliverable is an academic thesis document and a demonstrator pipeline. Weekly supervision meetings with Diptish Dey provide structured checkpoints.

6.4 Functional Requirements

The generator must produce typed event tuples conforming to the schema in Table 1: (user_id, session_id, action_type, item_id, delta_t). The five-action vocabulary of the Synerise dataset must be respected. Session structure must follow observed behavioral conventions: sessions begin on an arrival signal from the arrival simulator and terminate on an EOS token or a session length cap. The interface contract accepts the Arrival Simulator’s output as input and produces a log conforming to the schema. All generated sessions must pass through the validity layer before output.

6.5 Technical Requirements

The model must be trainable on a single GPU within the timeline. Vocabulary capping ensures softmax tractability by limiting the item space to the top- K most frequent items. Evaluation infrastructure must be deterministic: fixed random seeds, configuration-driven experiment runs, and all hyperparameters logged.

The evaluation pipeline must support reproducible execution across multiple random seeds. The generator must be deployable within the containerized simulation environment alongside the arrival simulator. Generation throughput must be sufficient to avoid becoming a bottleneck relative to the arrival simulator’s output rate. Throughput is reported as a diagnostic metric.

The evaluation protocol imposes the following measurement constraints. Fidelity is assessed via four metrics: Jensen-Shannon divergence over the action-type distribution (JSD), the two-sample Kolmogorov-Smirnov statistic over session length distributions (KS), L1 distance over the action bigram transition matrix, and L1 distance over the item bigram transition matrix restricted to item-bearing events (product_buy, add_to_cart, remove_from_cart). Validity is assessed via three constraint violation rates: illegal action transition rate, timestamp monotonicity violation rate, and purchase-without-prior-exposure rate. Both pre- and post-constraint-layer rates must be reported. Downstream utility is assessed via HR@10 and NDCG@10 under the TSTR paradigm. All metrics must be reported as mean and standard deviation across five random seeds. The transformer generator (TSTR-T) must outperform the Markov baseline (TSTR-M) on both HR@10 and NDCG@10 for the utility claim to be considered confirmed. Improvement on one metric

only is treated as inconclusive. The item bigram transition matrix must be computed and stored during preprocessing on the training split, restricted to top-K vocabulary items, for use as a fidelity reference during evaluation.

7 Methodology and System Design

7.1 Research Design Overview

The project follows a design-and-evaluate methodology across three phases: specification, modeling, and evaluation. The specification phase formalizes the interaction schema, simulator interfaces, and evaluation protocol. The modeling phase implements the Markov baseline and transformer generator. The evaluation phase runs the fidelity-validity-utility protocol with statistical reporting.

7.2 Interaction Schema and Interface Contract

The minimum generated event tuple is defined in Table 1. This schema formalizes what “interaction” means in the simulation context and determines what each prediction head of the generator must produce.

Table 1. Output log schema. Fields marked *generated* are produced directly by the generator; *derived* fields are computed post-hoc.

Field	Type	Description
user_id	integer	<i>Generated</i> . Identifies the synthetic user
session_id	integer	<i>Generated</i> . Groups events into one session
action_type	categorical	<i>Generated</i> . One of: page_visit, search_query, add_to_cart, remove_from_cart, product_buy
item_id	integer	<i>Generated</i> . The item involved in the event. Null for search_query events, which do not target a specific item.
delta_t	float	<i>Generated</i> . Inter-event time in seconds
event_time	timestamp	<i>Derived</i> . Cumulative delta_t from session start

The item_id field is nullable; the item prediction head is suppressed for search_query events and no item target is produced or evaluated at those positions.

Session boundaries are determined by an EOS token predicted by the generator, subject to a maximum session length cap. The generator accepts the output of the arrival simulator as the input and produces the above schema. Context features such as category and price are excluded from the minimum contract and identified as future work.

7.3 System Architecture

The simulator pipeline is:

arrival input → interaction generator → hard-constraint validation → output logging

Multi-session user state.

This work adopts a session-based generation approach with cross-session user history context. Each session is generated as an independent sequence, but the generator conditions on the user’s prior session history up to a window of H events, capturing behavioral evolution across visits.

Transformer backbone. At each time step t , three prediction heads share the transformer hidden state h_t :

- $p(\text{action}_t \mid h_t)$: softmax over 6 classes — the 5 action types plus an EOS token signaling session termination;

- $p(\text{item}_t \mid \text{action}_t, h_t)$: softmax over the item vocabulary, conditioned on the predicted action type;
- $p(\delta_t \mid \text{action}_t, \text{item}_t, h_t)$: temporal head, either via discretized log-scale time bins.

Input representation at each sequence position is the sum of component embeddings:

$$\mathbf{x}_t = \mathbf{e}_{\text{item}} + \mathbf{e}_{\text{action}} + \mathbf{e}_{\delta_t} + \mathbf{e}_{\text{pos}}$$

Training uses teacher forcing with ground-truth event tuples at each position. The EOS token is the final target of every training session, so the model learns session termination from real boundaries. All three heads are trained jointly; the training loss is cross-entropy over action type (including EOS), cross-entropy over item, and temporal loss over discretized bins.

Factored prediction. A flat action–item composite vocabulary would be intractably large. Factored heads keep each softmax computationally tractable while preserving action-item joint modeling through the conditional head structure.

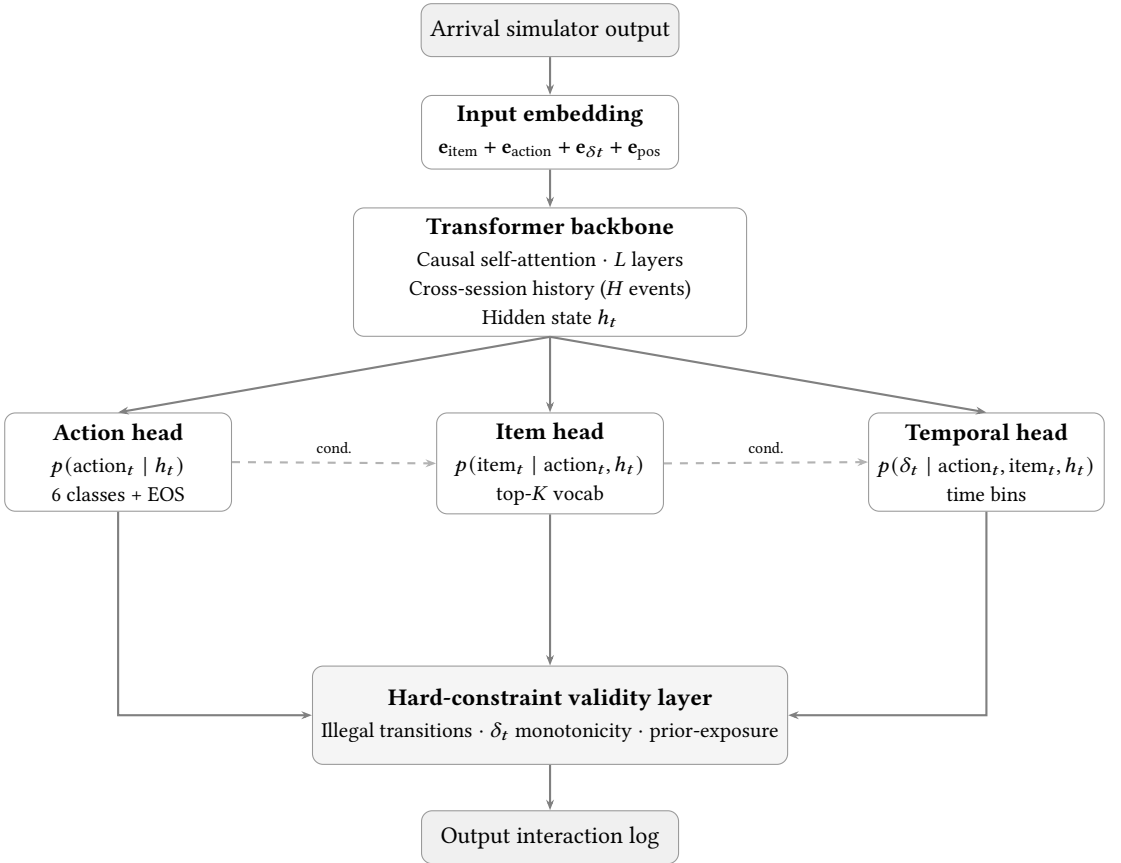


Fig. 1. System architecture. The transformer backbone produces a shared hidden state h_t at each step, from which three factored heads jointly generate the action type, item, and inter-event time. The dashed arrows indicate conditioning: the item head conditions on the predicted action, and the temporal head conditions on both. All sessions pass through the hard-constraint validity layer before output.

7.4 Hard Constraint Validity Layer

Generated sessions pass through explicit validity checks before output:

- (1) **Illegal action transitions:** action bigrams not observed in training data are flagged and counted.
- (2) **Timestamp monotonicity:** events where $\delta_t < 0$ are flagged.
- (3) **Purchase-without-prior-exposure:** a product_buy event with no preceding page_visit or add_to_cart within the session is flagged.

Both pre-constraint and post-constraint statistics are reported, making the cost of constraint repair independently measurable.

7.5 Comparison Portfolio

Three conditions are evaluated:

- **TRTR** (Train on Real, Test on Real): GRU4Rec trained on the real training split; serves as the interpretive upper bound.
- **TSTR-M** (Train on Markov-Synthetic, Test on Real): A first-order Markov chain over action types, with item selection drawn from the empirical popularity distribution of the training split. This baseline has no action-item coherence: items are sampled independently of the action type at the same event. Any coherence advantage in TSTR-T results reflects a genuine architectural difference between the two generators. This baseline mirrors the current interaction generator in the simulation pipeline, making the comparison directly relevant to the improvement objective of this thesis.
- **TSTR-T** (Train on Transformer-Synthetic, Test on Real): the proposed multi-head autoregressive transformer.

7.6 Reliability and Validity

Reliability. Test-retest reliability is operationalized through three-seed variance reporting. Running the same generator with five different random seeds plays the role of repeated measurements: low variance across seeds indicates that the measurement is stable and not an artifact of random initialization. This approach follows the repeated-run practice recommended by T-RECS [5].

Validity. The three fidelity metrics each target a distinct property of the generated sessions. JSD over action-type distributions captures whether the generator produces the right behavioral mix at the marginal level [19]. The KS test on session length checks whether engagement patterns, how long a session runs, match the real distribution. L1 distance on the action bigram transition matrix goes one level deeper, checking whether the sequential structure of actions is preserved, not just their overall frequency. A generator could pass JSD while failing L1 if it produces the right action mix in the wrong order.

A fourth metric addresses item-level sequential coherence: L1 distance on the item bigram transition matrix, computed over the subsequence of item-bearing events (product_buy, add_to_cart, remove_from_cart). page_visit and search_query events are excluded as they carry no sku field in the Synerise schema. Item bigrams are computed over the top-K capped vocabulary only. Transitions involving UNK tokens are excluded from the matrix. This metric captures whether the generator reproduces realistic item co-occurrence patterns within sessions, for example, whether items added to cart are plausibly related to previously viewed items.

For downstream utility, HR@10 and NDCG@10 are the standard benchmarks in sequential RS research, used by SASRec, BERT4Rec, and GRU4Rec [15–17]. Their use here follows the TSTR evaluation paradigm established in Jordon et al. [11].

Ecological validity.

Offline evaluation results are bound to static log fidelity and training-set utility; they do not support claims about generator performance under live deployment with real feedback dynamics.

Content validity. The five-action vocabulary covers the complete observed action space of the Synerise dataset. No action type is excluded from the evaluation.

8 Data Strategy**8.1 Dataset**

The primary dataset is the Synerise RecSys 2025 dataset: 225M events over 168 days with a five-type action vocabulary (page_visit, search_query, add_to_cart, remove_from_cart, product_buy). Its rich action schema and temporal span make it well-suited for evaluating constraint-aware interaction generation. The dataset originates from a retail e-commerce platform; the behavioral funnel it encodes, browsing, search, cart management, and purchase, is domain-specific. The interaction schema, hard constraints, and validity rules defined in this thesis represent retail conventions and would require re-derivation from domain-appropriate data before being applied to other contexts such as news or media.

8.2 Data Preparation Pipeline

- (1) **Schema normalization:** merge the Synerise action parquets and join product_properties for category and price fields.
- (2) **Sessionization:** a 30-minute inactivity threshold is applied for session segmentation, following common practice in session-based RS research. The chosen threshold is verified against the inter-session gap distribution during preprocessing and reported.
- (3) **Time-aware split:** train June 23 – November 15, 2022; validation November 15 – December 1; test December 1 – December 8, capturing behaviorally distinct high-activity periods in held-out data.
- (4) **Vocabulary capping:** item vocabulary is capped to the top- K most frequent items, with K chosen to achieve sufficient event coverage on the training split; remaining items map to a shared UNK token. The UNK replacement rate is reported as a generation diagnostic.
- (5) **Constraint rule extraction:** legal action bigrams and exposure prerequisites are extracted from training data and used by the hard-constraint layer.

The dataset consists of anonymized behavioral logs; no personally identifiable information is processed.

9 Evaluation Protocol**9.1 Setup**

Evaluation covers three dimensions: fidelity (do generated sessions look like real ones?), validity (do they respect behavioral constraints?), and downstream utility (do they train a better recommender?).

All metrics are reported across five random seeds. Results are presented as mean and standard deviation; no formal threshold-passing criteria are applied.

The evaluation design is grounded in the reliability and validity criteria established in and detailed in the Reliability and Validity subsection of the Methodology. Three-seed reporting operationalizes test-retest reliability.

9.2 Fidelity

Four distributional metrics compare generated sessions against the real training data:

- **JSD(action distribution):** Jensen-Shannon divergence between real and generated action type frequencies.
- **KS(session length):** two-sample Kolmogorov-Smirnov test on session length distributions.
- **L1(action transition matrix):** L1 distance between real and generated action bigram transition matrices.
- **L1(item transition matrix):** L1 distance between real and generated item bigram transition matrices, computed over item-bearing events only (`product_buy`, `add_to_cart`, `remove_from_cart`). Restricted to the top- K capped vocabulary. Captures within-session item-level sequential coherence.

Both generators are evaluated on the same metrics, making the comparison between TSTR-M and TSTR-T direct.

9.3 Validity

Generated sessions are checked against three hard behavioral constraints extracted from the training data:

- Invalid action transition rate (bigrams not observed in training data).
- Timestamp monotonicity violation rate ($\delta_t < 0$).
- Purchase-without-prior-exposure rate.

Violation rates are reported before and after the hard-constraint layer, making the repair cost visible.

9.4 Downstream Utility

Utility follows the Train-Synthetic-Test-Real (TSTR) paradigm:

- (1) Train each generator on the training split.
- (2) Generate a synthetic training set of equal session count.
- (3) Train GRU4Rec on the synthetic set; evaluate on the real held-out test split.
- (4) Compare TSTR-T against TSTR-M on HR@10 and NDCG@10.

A Train-Real-Test-Real (TRTR) run provides an interpretive reference point. The primary claim is directional, meaning that TSTR-T should outperform TSTR-M on both HR@10 and NDCG@10. Improvement on one metric only is reported but treated as inconclusive.

10 Planning

Table 2 shows the phase-based schedule.

11 Risks, Mitigations, and Scope Boundaries

Risk 1: Metric gaming or distributional collapse. A generator that overspecializes on a single evaluation signal may pass individual metrics while failing distributional realism overall. Mitigation: the multi-axis evaluation protocol (fidelity, validity, and utility jointly) prevents single-axis gaming, and the exact duplicate session rate provides an independent collapse signal.

Risk 2: Reproducibility drift. Re-running experiments with slightly different configurations can produce inconsistent results. Mitigation: fixed random seeds, deterministic preprocessing pipelines, configuration-driven experiment runs with logged hyperparameters, and variance reporting across all seeds.

Risk 3: Preprocessing cost at scale. At 225M events, sessionization and vocabulary capping may consume significant calendar time. Mitigation: preprocessing begins on a small sample to validate the pipeline end-to-end before committing to the full dataset.

Table 2. Implementation and writing schedule

Phase	Period	Activity
Literature & Specification	March (weeks 1–2)	Supervisor kick-off; literature study and schema finalization
Data Preparation	March – April (weeks 3–4)	Data collection and preprocessing; pipeline validation
Baseline & Pilot	April (week 5)	Baseline implementation and initial pipeline validation
Model Development	April – May (weeks 6–8)	Model design, training, and initial evaluation
Evaluation	May (week 9)	Full evaluation run; Portfolio 70% due 11 May
Extensions & Writing	May (weeks 10–12)	Optional extensions; results and discussion
Review & Final	June	Thesis review, corrections, and formatting; Portfolio 100% due 12 June

Risk 4: Ecological validity gap. Offline evaluation results are bounded to static log fidelity and training-set utility and do not predict generator performance under live deployment with real feedback dynamics. Mitigation: all claims are explicitly scoped to offline evaluation; live deployment is out of scope.

Risk 5: Generation throughput bottleneck. At inference time, autoregressive generation is inherently sequential, each token conditions on all prior tokens, which limits throughput compared to parallel generation approaches. If the generator cannot keep pace with the arrival simulator at scale, the pipeline will stall. Mitigation: generation throughput is measured and reported as a diagnostic metric; batched generation is identified as a fallback if single-session throughput proves insufficient at scale.

Scope boundaries. In scope: the interaction generator, evaluation protocol, demonstrator pipeline, and thesis document. Out of scope: context feature integration, RNN/GRU comparison, production deployment, live A/B testing, long-horizon feedback-loop simulation, GAN-based and LLM-based generation, and online evaluation. These are identified as natural directions for future work.

12 Expected Deliverables

0. **Literature Study Report:** a standalone document synthesizing the reviewed corpus on interaction definitions, modeling approaches, simulation techniques, and evaluation methodology. This report feeds the design decisions in all subsequent deliverables.
- (1) **Interaction schema and simulator interface contract:** event-level typed tuple definition with field specifications, action type vocabulary, and session boundary rules; documented interface from the arrival simulator.
- (2) **Transformer-based interaction generator:** multi-head autoregressive transformer, integrated with the hard-constraint validity layer and the process-based simulation pipeline.

- (3) **Baseline comparison results:** Markov chain vs transformer generator evaluated across all three axes (fidelity, validity, utility), with results reported as mean and standard deviation across five random seeds.
- (4) **Evaluation report:** results across fidelity, validity, and utility interpreted jointly; includes discussion of failure modes where results fall short of expectations.
- (5) **Demonstrator pipeline:** a configuration-driven script that accepts a generator type and random seed, runs the full TSTR evaluation loop, and outputs a structured results table. A sample run on a held-out subset of the Synerise dataset is included as proof of function.

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